

CFD-HW4

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Outline

In this assignment, I was tasked with creating piping geometry consistent with the provided (textbook) dimensions, following by meshing, simulating (fluid flow & pressure drop), and finally comparing the results to the textbook's. It will be assumed the fluid of study is water.

Geometry

The provided geometry depicted a hollow pipe formed by two connected cylinders. The first cylinder has a diameter of $D_1 = 0.0191$ m. The second (smaller) cylinder has a diameter of $D_2 = 0.0102$ m. Both have a length of $L = 0.3$ m. They are connected (centered) at their respective faces/caps.

The preliminary CAD model was made in Autodesk Inventor, and was subsequently exported into Ansys's Discovery tool in the workbench. Within Discovery, the pipe's body was split into two distinct sections for meshing purposes.

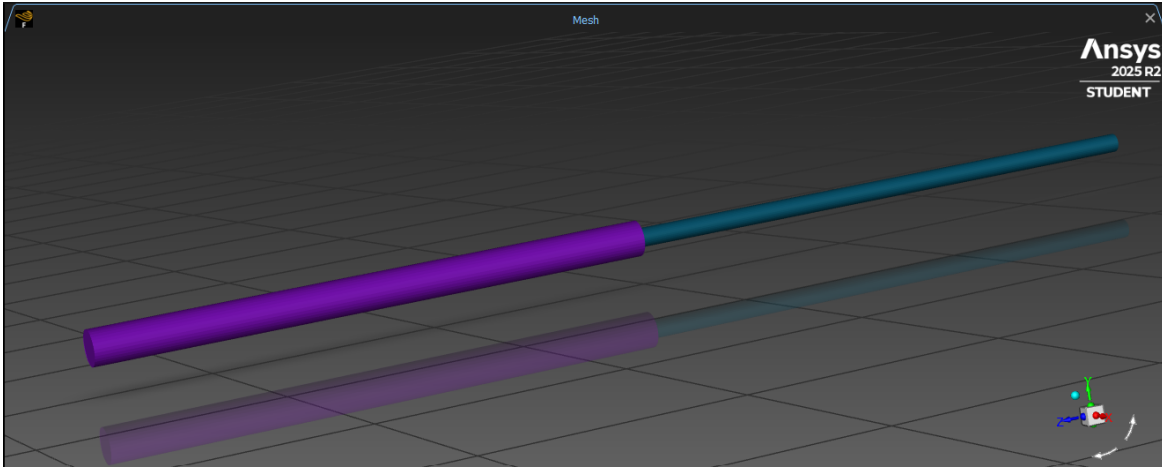


Figure 1: Preliminary Geometry Work

Preliminary Calculations

Before performing the meshing and CFD analysis, a baseline calculation ought to be performed. To start, water has a density $\rho = 1000 \text{ kg/m}^3$ and viscosity $\mu = 0.001 \text{ Pa} \cdot \text{s}$. Because the initial Reynolds number is $\text{Re}_1 = 372 < 2300$, the flow is laminar. The respective friction factor is roughly $f_1 \approx 64/\text{Re}_1 \approx 0.172$. The inlet speed at the inlet can be calculated as such:

$$V_1 = \frac{\mu \text{Re}_1}{\rho D_1} = \frac{(0.001 \text{ Pa} \cdot \text{s})(372)}{(1000 \text{ kg/m}^3)(0.3 \text{ m})} \approx 0.00124 \text{ m/s}$$

By conservation of mass, $\rho V_1 A_1 = \rho V_2 A_2$, where A_1 and A_2 are the inlet area and contraction interface area. Therefore, it follows that:

$$V_2 = \frac{A_1}{A_2} V_1 = \left(\frac{D_1}{D_2} \right)^2 V_1 \approx \left(\frac{0.0191}{0.0102} \right)^2 (0.00124 \text{ m/s}) \approx 0.00435 \text{ m/s}$$

The small pipe is expected to have the following Reynolds number as given below:

$$\text{Re}_2 = \frac{\rho V_2 D_2}{\mu} \approx \frac{(1000 \text{ kg/m}^3)(0.00435 \text{ m/s})(0.0102 \text{ m})}{0.001 \text{ Pa} \cdot \text{s}} \approx 44.35 < 2300 \implies f_2 \approx 64/\text{Re}_2 \approx 1.443$$

The net minor pressure drop (due to friction) can now be computed as follows:

$$\Delta P_{\text{minor}} = f_1 \frac{L_1}{D_1} \frac{\rho V_1^2}{2} + f_2 \frac{L_2}{D_2} \frac{\rho V_2^2}{2} \approx (0.172) \frac{0.3}{0.0191} \frac{(1000)(0.00124)^2}{2} + (1.443) \frac{0.3}{0.0102} \frac{(1000)(0.00435)^2}{2} \text{ Pa} \approx 0.403 \text{ Pa}$$

The net major pressure drop (due to the sudden change in geometry), and thus the overall net pressure drop, can now be computed as such (with $K_c \approx 0.4$):

$$\Delta P_{\text{major}} = K_c \frac{\rho V_2^2}{2} \approx (0.4) \frac{(1000 \text{ kg/m}^3)(0.00435 \text{ m/s})^2}{2} \approx 0.004 \text{ Pa} \implies \Delta P_{\text{net}} = \Delta P_{\text{minor}} + \Delta P_{\text{major}} \approx 0.41 \text{ Pa}$$

The First Mesh

For the first mesh, I used all the default settings with a few exceptions. First, I shrank the maximum cell size to 0.0025 m. Second, I had to make sure the two regions (cylinders) shared the overall topology with their interfaces being registered as internal. Lastly, I used the Polyhexcore mesh filler.

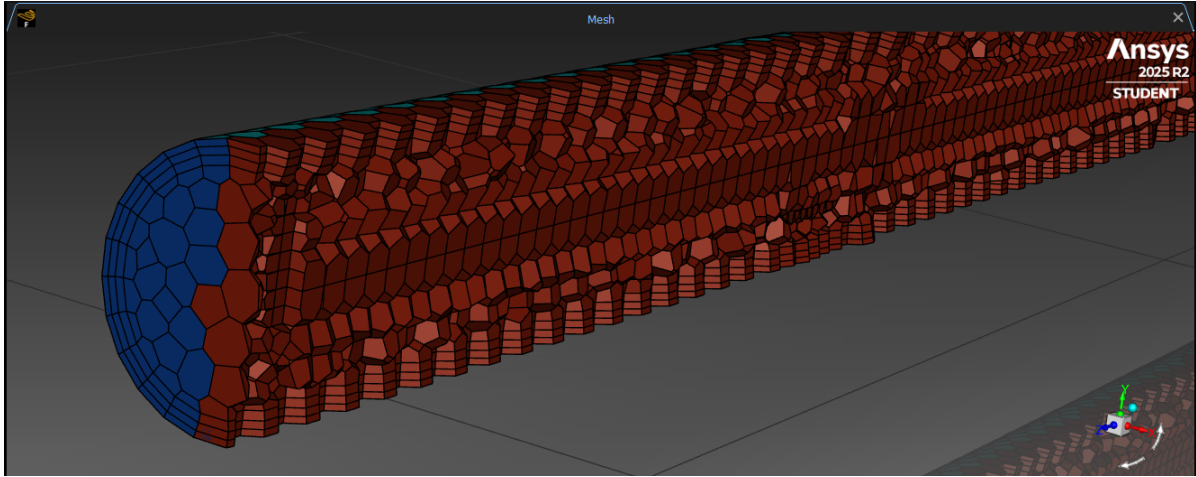


Figure 2: The First Mesh

Upon implementing the mesh into Fluent, the respective fluid (water) was assigned to the cell zones of interest and the boundary conditions were executed. To be precise, the velocity was initialized at the inlet (0.00124 m/s), and the turbulence settings were fixed at the inlet/outlet. The k-omega (SST) model was engaged, and the model was ready for initialization.

Below is a residual plot, which demonstrates convergence. It is also accompanied by the average pressure reading (for each iteration) at the inlet/outlet, which also* demonstrates convergence.



Figure 3: Residuals with Respect to the First Mesh

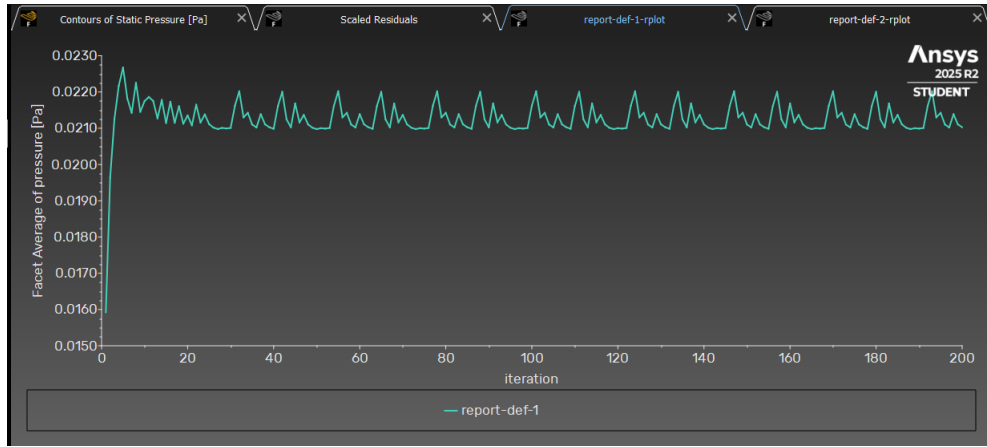


Figure 4: Inlet Average Pressure with Respect to the First Mesh

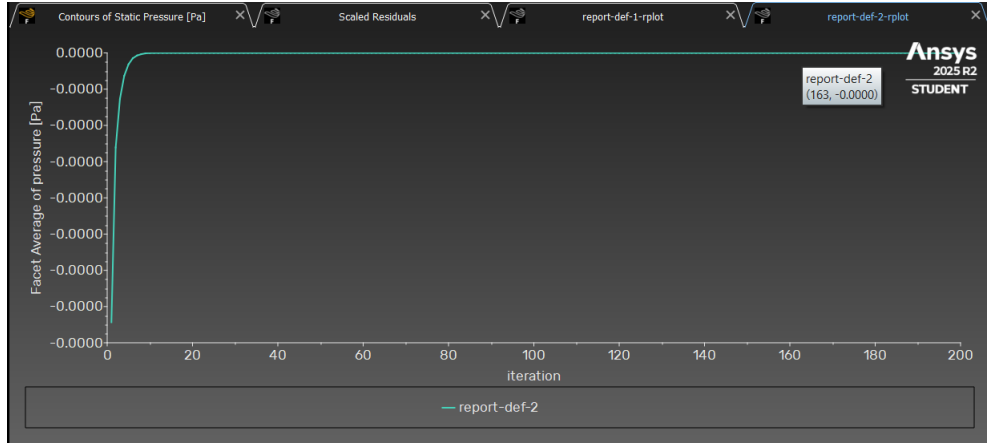


Figure 5: Outlet Average Pressure with Respect to the First Mesh

Upon inspection, the computed pressure drop across the pipe (end to end) was determined to be ≈ 0.02 Pa, which is an order of magnitude off from the preliminary calculations. Not a great sign.

At this point, the velocity profiles at various locations in the pipe will be compared to that in the textbook. This will be achieved by constructing planes and lines in the geometry from which many plots can be made. The origin of the coordinate system is at the center of the contraction interface. The (new) x-axis is normal to the interface, oriented positively with respect to the initial fluid velocity. The variable r denotes the radial displacement from the (new) x-axis. Note that u_{ij} denotes the respective normalized speed (i.e. $u/\bar{u}$).

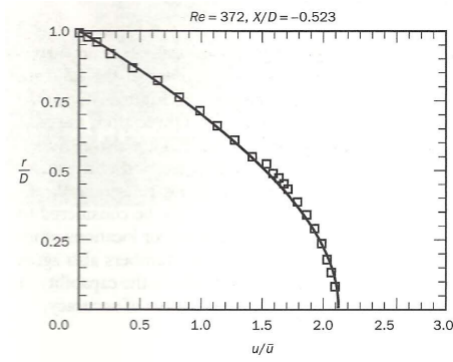


Figure 6.11a: TB Speed Profile: $x/D_1 = -0.523$

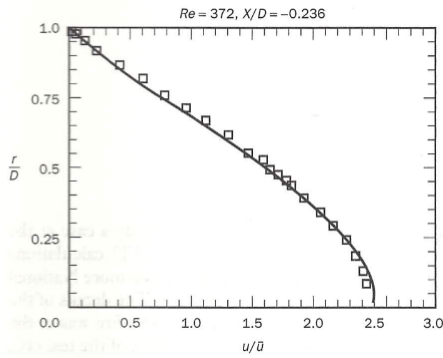


Figure 6.21a: TB Speed Profile: $x/D_1 = -0.236$

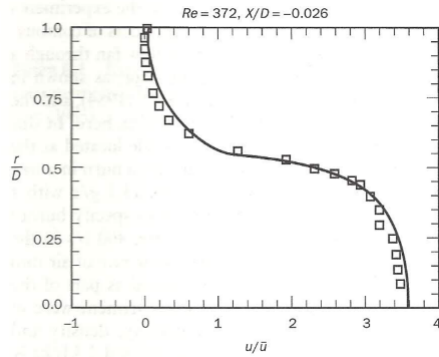


Figure 6.31a: TB Speed Profile: $x/D_1 = -0.026$

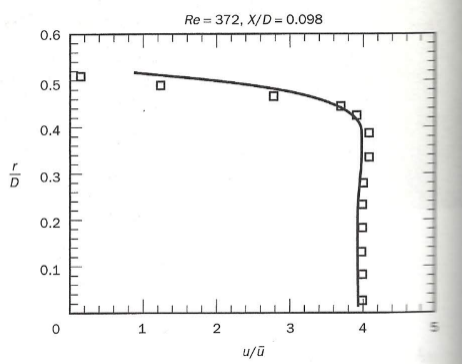


Figure 6.12a: TB Speed Profile: $x/D_1 = 0.098$

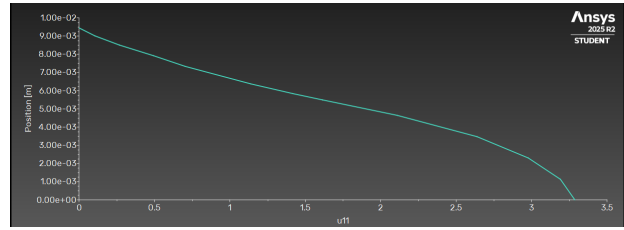


Figure 6.11b: Calc. Speed Profile: $x/D_1 = -0.523$

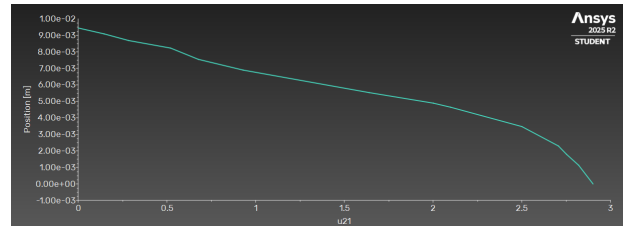


Figure 6.21b: Calc. Speed Profile: $x/D_1 = -0.236$

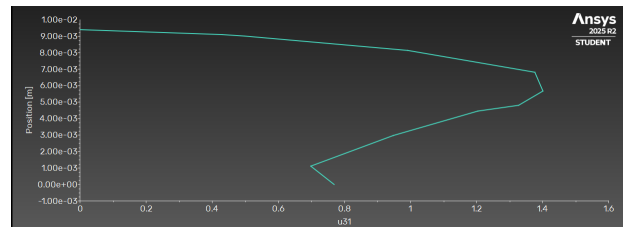


Figure 6.31b: Calc. Speed Profile: $x/D_1 = -0.026$

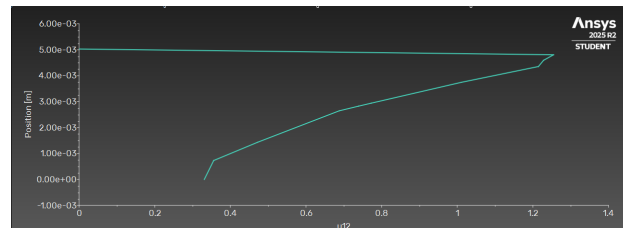


Figure 6.12b: Calc. Speed Profile: $x/D_1 = 0.098$

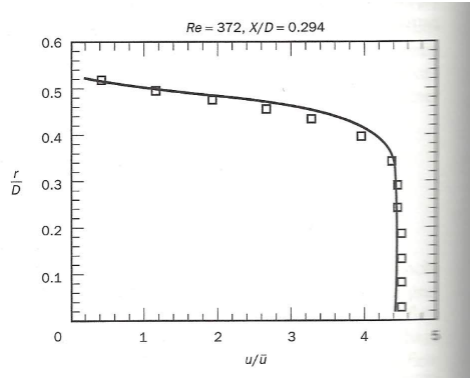


Figure 6.22a: TB Speed Profile: $x/D_1 = 0.294$

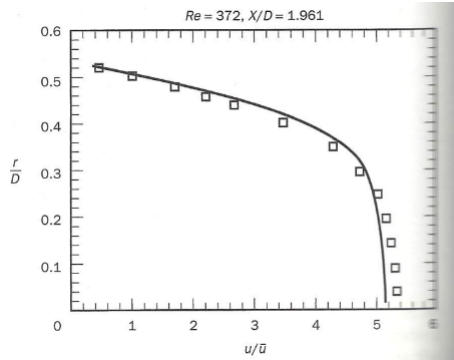


Figure 6.32a: TB Speed Profile: $x/D_1 = 1.961$

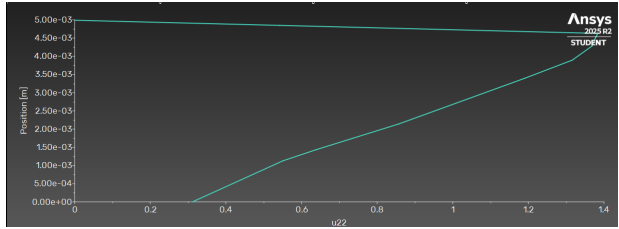


Figure 6.22b: Calc. Speed Profile: $x/D_1 = 0.294$

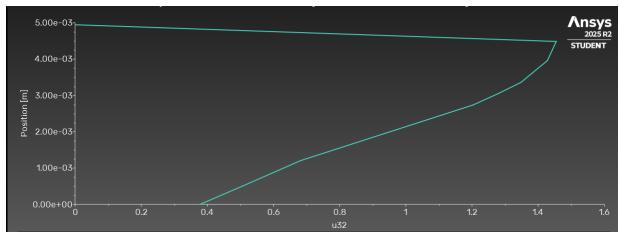


Figure 6.32b: Calc. Speed Profile: $x/D_1 = 1.961$

Aside from the fact that I didn't format the vertical axis correctly (i.e normalize position), the results obviously do not match up in the slightest; at least characteristically. Hopefully, by implementing a superior meshing scheme, this can be remedied!

It is at this point that my local (fluent) files got corrupted after my computer crashed, and I lost all of my progress. Guess I need to start from scratch!

The Second Mesh

This time I used a much denser mesh in conjunction with multi zone, and the interface connect feature in the mesher. I wanted to make sure the interface had enough cells to capture the physics properly.

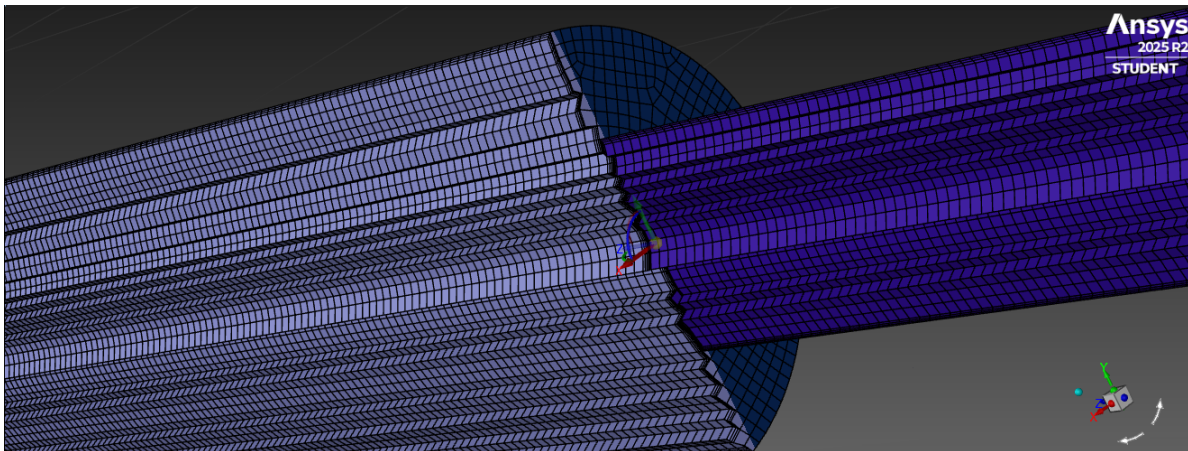


Figure 7: The Second Mesh

Upon implementing the mesh into Fluent, the respective fluid (water) was assigned to the cell zones of interest and the boundary conditions were executed. To be precise, the velocity was initialized at the inlet (0.00124 m/s), and the turbulence settings were fixed at the inlet/outlet. The k-omega (SST) model was engaged, and the model was ready for initialization.

Because I suck at meshing, I ended up generating A LOT of cells to chase the results I was after. That inefficiency gave me the extra luxury of waiting 35 minutes for Navier-Stokes to iterate 150 times over the entire mesh. Many lessons have been learned while going through with this assignment...

Below is a residual plot, which demonstrates convergence. It is also accompanied by the average pressure reading (for each iteration) at the inlet/outlet, which also* demonstrates convergence.

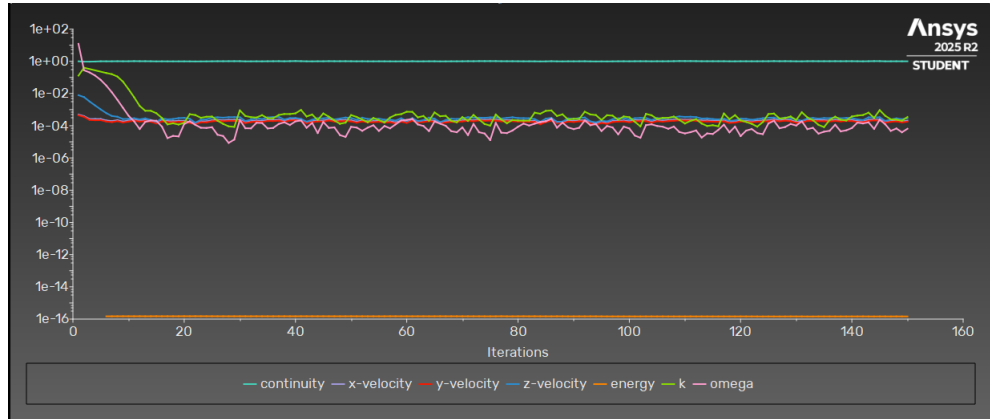


Figure 8: Residuals with Respect to the Second Mesh

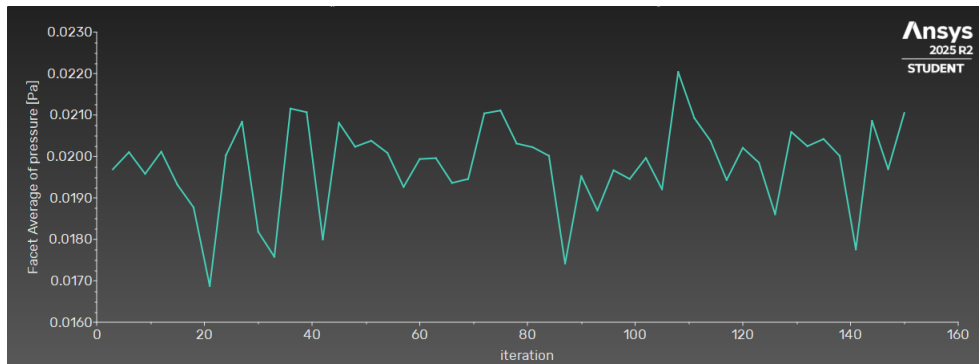


Figure 9: Inlet Average Pressure with Respect to the Second Mesh

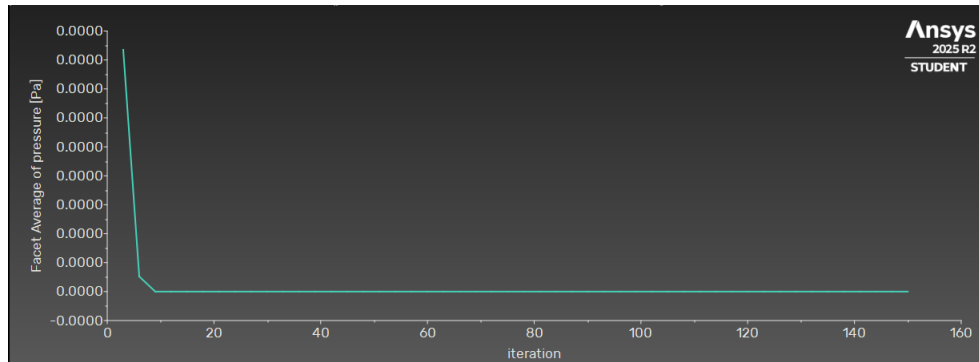


Figure 10: Outlet Average Pressure with Respect to the Second Mesh

Here too, the pressure drop is off by an order of magnitude from the preliminary calculation; almost exactly by the same amount. This suggests the calculation is off.

Again, the velocity profiles at various locations in the pipe will be compared to that in the textbook. The origin of the coordinate system is at the center of the contraction interface. The (new) x-axis is normal to the interface, oriented positively with respect to the initial fluid velocity. The variable r denotes the radial displacement from the (new) x-axis. Note that u_{ij} denotes the respective normalized speed (i.e. $u/\bar{u}$). Also, now xperd is short for x per unit D_1 (i.e. r/D) in the old coordinate system.

Just to be absolutely clear, I painfully defined 6 offset planes in just the right locations, and I constructed 6 line segments to sit right on top of them. From there, I recorded the average fluid speed over each plane, and then I defined the 6 custom variables u_{ij} so I could compare the results to the textbook. Also note my naming/indexing convention originated from the matrix-like arrangement (row/column) of the printed textbook plots, which I must admit was a mistake to adhere to.

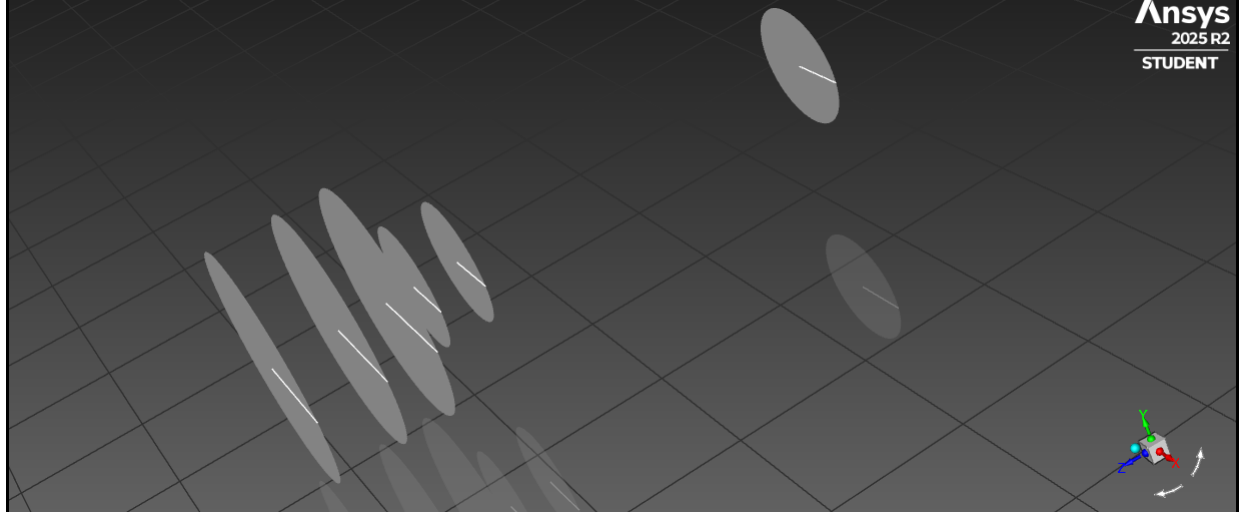


Figure 11: Preliminary Plotting Work (pain)

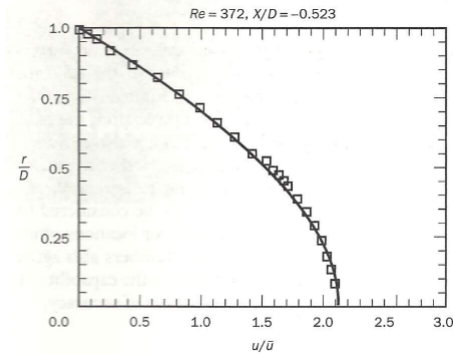


Figure 12.11a: TB Speed Profile: $x/D_1 = -0.523$

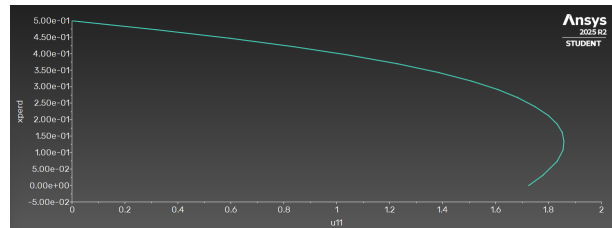


Figure 12.11b: Calc. Speed Profile: $x/D_1 = -0.523$

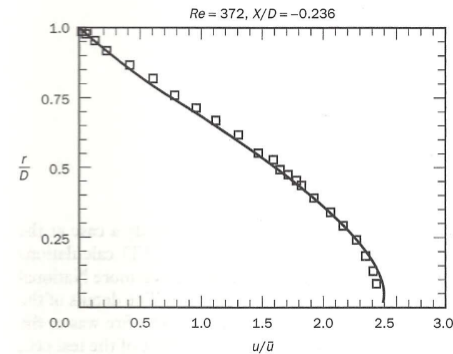


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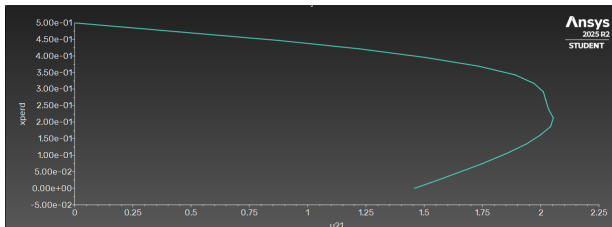


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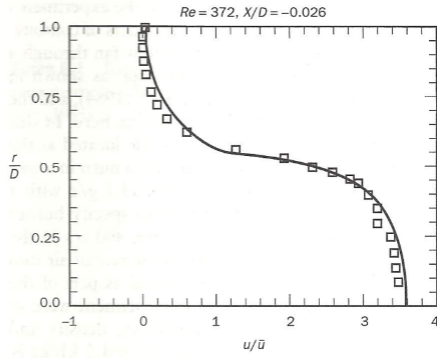


Figure 12.31a: TB Speed Profile: $x/D_1 = -0.026$

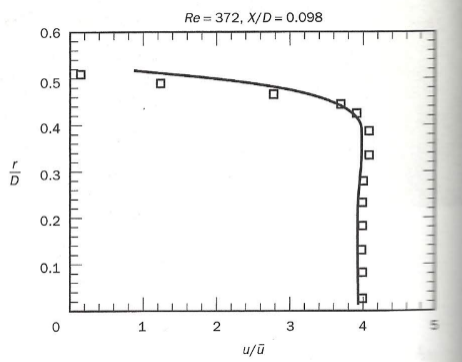


Figure 12.12a: TB Speed Profile: $x/D_1 = 0.098$

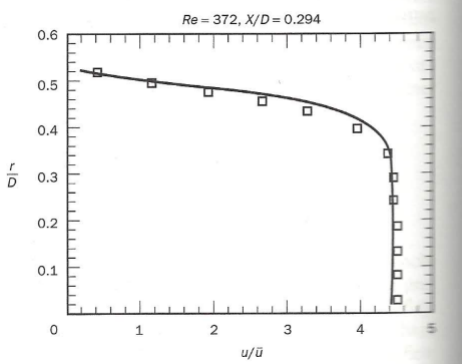


Figure 12.22a: TB Speed Profile: $x/D_1 = 0.294$

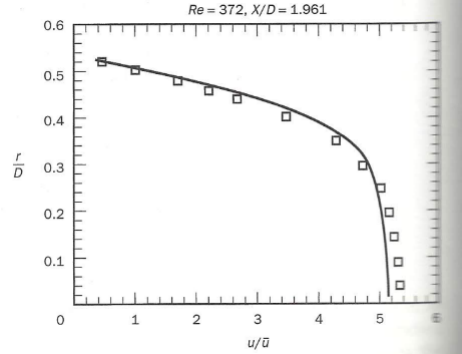


Figure 12.32a: TB Speed Profile: $x/D_1 = 1.961$

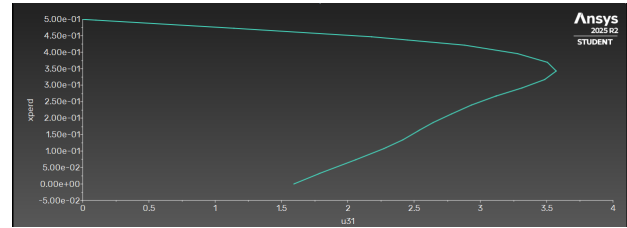


Figure 12.31b: Calc. Speed Profile: $x/D_1 = -0.026$

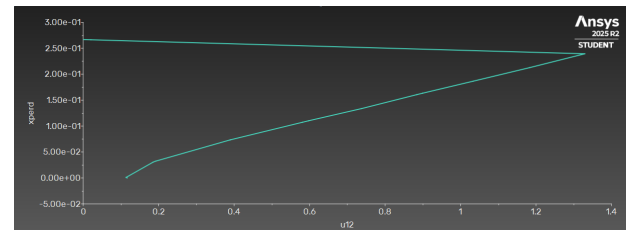


Figure 12.12b: Calc. Speed Profile: $x/D_1 = 0.098$

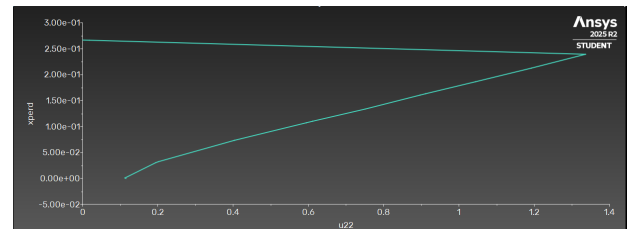


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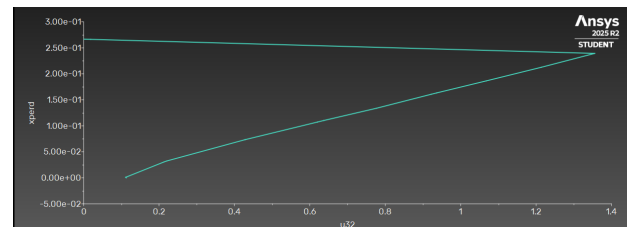


Figure 12.32b: Calc. Speed Profile: $x/D_1 = 1.961$

It goes without saying this was an absolute failure. The first thing that catches my eye now that I've brought myself to normalize the position is that my geometry (diameter) is off by a factor of two because I can't read instructions. Regardless, the shape of the speed profiles are alien relative to the ones in the textbook. Not the best look...

On the bright side, the results from meshes 1 and 2 are similar, which indicates both simulations were converging to (possibly) the same physical configuration... Just not the one in the textbook.

Conclusionary Remarks

Overall, the sought after speed profiles presented by the textbook were not recreated due to a plethora of shortcomings. However, many lessons were picked up on the journey to failure, which will hopefully not be in vein for the sake of my mental state.